FROM THE ANALYST'S COUCH

Lysosomal storage diseases market

Yaron Werber



'Jeff Gordon Flames Sofa'. Image courtesy of Racing Furniture www.racingfurniture.com

Interest in lysosomal storage diseases (LSDs) has increased significantly, because several new therapeutics received FDA approval in 2003. Though long overlooked by most major pharmaceutical companies, LSDs present attractive opportunities for clinical development because of their well-understood pathology, straightforward path of intervention (that is, enzyme replacement), accelerated development timelines, Orphan Drug exclusivity (seven years in the United States and ten years in Europe), and premium pricing ($170,000 per patient per year on average). Approximately eight LSDs will probably prove amenable to therapeutic intervention by 2008, and the combined market opportunity for therapeutics that treat these diseases could exceed US $1.9 billion in that year.

Still growing after all these years

Gaucher's disease is caused by a deficiency of glucocerebrosidase (GCR) and was the first disease in which enzyme-replacement therapy became available to patients. Genzyme originally launched Ceredase (placenta-derived GCR) in 1991, which was followed by the approval of Cerezyme (recombinant GCR) in 1994. Since that time, sales have grown steadily at a 28% compound annual growth rate and are expected to reach US $732 million in 2003. At present, of the estimated 5,000–6,000 patients afflicted with the disease worldwide, approximately 3,800 are receiving therapy. Although the market has matured, average per-patient doses (weight-based dosing) have increased and further market penetration through ongoing efforts to identify patients have catalysed growth. Gaucher's disease has proved to be particularly successful from a commercial standpoint due to Cerezyme's strong ability to improve the pathophysiology of the disease, long patient lifespan and large numbers of afflicted patients. It remains to be seen whether other enzyme-replacement therapies can duplicate its economic success.

Fabry's bigger than Gaucher's?

Fabry's disease is an X-linked disorder caused by a deficiency of α-galactosidase. The resulting accumulation of globotriaosylceramide (GB_3) in kidney, heart and neural tissue is responsible for the slow progressive course of the disease. Patients typically succumb to cardiac arrest, stroke and renal failure between 40 and 50 years of age. In 2001, the European Union approved Genzyme's Fabrazyme and Transkaryotic Therapies' (TKT) Replagal. Both drugs are formulations of α-galactosidase, but are produced in different cell lines. In the United States, Genzyme beat TKT and garnered exclusive US approval with the seven-year Orphan Drug exclusivity due to a superior clinical development programme. Experts believe that 4,000–5,000 patients suffer from the disease worldwide. However, that number might be underestimated and sales of these drugs might eclipse those of Cerezyme in the long-term.

MPS-I: a more challenging market?

Mucopolysaccharidosis I (MPS-I) is caused by an autosomal recessive deficiency of α-L-iduronidase, and is seen in three forms, which are not mutually exclusive or distinct from each other, but which do display different severities of common symptoms. Hurler's syndrome, the most severe form, is diagnosed earliest in life (within the first 18 months) and carries the poorest prognosis. Patients experience mental retardation, reduced cardiopulmonary performance and severe joint restriction. Patients afflicted with Hurler–Scheie syndrome, an intermediate form, have many of the same symptoms, but show a slower rate of deterioration. Scheie's syndrome patients, the mildest form, have less extensive disease and can live a normal lifespan. Although an estimated 3,000 patients suffer from MPS-I worldwide, an estimated 60–70% of patients suffer from Hurler's syndrome and display mental retardation. Bone-marrow transplantation (BMT) has shown promising results in these patients, often leading to improved survival. Because of the increasing adoption of BMT, the role of the enzyme-replacement therapy Aldurazyme (Genzyme/BioMarin) is unclear. Aldurazyme was launched in the United States and Europe in 2003.

Pompe's: largest commercial opportunity

Pompe's disease, which afflicts between 5,000–10,000 patients worldwide, is primarily a disorder of cardiac and skeletal muscle. The infantile-onset form of Pompe displays an aggressive course, in which patients are typically diagnosed within the first 12 months of life and die by two to three years of age of severe cardiomyopathy. Adult-onset patients typically display skeletal myopathy and might live into adulthood depending on the disease severity. Given that the disease affects the muscle tissue, the required enzyme dosing is 20–40 times that of Fabrazyme in Fabry's disease, in which the disorder is more limited to the kidney. Genzyme has launched multiple Phase III studies and expects to file for regulatory approval by year-end 2004. Initially, Genzyme will focus on the infantile-onset form and has designed the trial to show a survival benefit. A number of earlier Phase I studies have outlined the ability of Myozyme (Genzyme), recombinant human α-glucosidase, to increase survival in afflicted infants. To facilitate early diagnosis, Genzyme's Genetics business unit is developing a screening diagnostic to test all births for the disorder.

Opportunities exist in small markets

TKT is developing iduronate-2-sulphatase (I2S) to treat Hunter's syndrome (MPS-II), which is caused by an X-linked deficiency of I2S. Patients suffering from the disorder manifest similar symptoms to MPS-I patients, but with differing severities. TKT initiated the pivotal trial of I2S in 2003 and expects to file for marketing approval in early 2005. Nearly all 2,000 patients with Hunter's syndrome have been diagnosed, thereby facilitating marketing efforts. TKT will seek to commercialize I2S by building a US sales force and will also employ its European sales force. TKT has partnered with Genzyme to leverage its strong commercialization expertise in Japan.

BioMarin's Aryplase has recently entered Phase III testing to treat Maroteaux–Lamy syndrome (MPS-VI), an autosomal recessive disease. Patients are typically diagnosed in the first six to twenty-four months of life and die of cardiomyopathy in their teens. BioMarin expects to file for marketing approval by year-end 2004. Although Aryplase will probably yield encouraging clinical results, the small market size might translate into a challenging commercialization effort. BioMarin intends to seek a partner for ex-US distribution of the drug. In the United States, the company will probably lead the commercialization efforts. ▶

LYSOSOMAL STORAGE DISEASES MARKET | MARKET INDICATORS

Lysosomal storage diseases (LSDs) affect a significant number of individuals (FIG. 1) and some can be treated with enzyme-replacement therapies (TABLE 1). However, because enzymes cannot cross the blood–brain barrier, replacement therapeutics are unable to address the central nervous system manifestations of the disorders. Four drugs for LSDs are presently in preclinical testing (TABLE 2) and most of these are expected to enter Phase I trials in 2004. Three of the four are enzyme-replacement therapies, including TKT's competitor to Genzyme's Cerezyme. Although Actelion's Zavesca, an oral small molecule, has recently garnered approval for Gaucher's disease in patients who cannot tolerate enzyme replacement, the drug is toxic and has a modest clinical benefit. Genzyme's small-molecule P4 analogue (GENZ-112638) is many times more potent than Zavesca and displays a more attractive therapeutic index. In addition, it has shown preclinical activity in Tay–Sachs disease and Sandhoff syndrome. Although enzyme replacement will continue to be the mainstay of therapy, oral small molecules can provide a tangible convenience advantage by replacing the weekly intravenous infusion.

Yaron Werber, M.D., is a Vice President biotech analyst at SG Cowen Securities, 1221 Avenue of the Americas, New York, New York 10020, USA.
e-mail: Yaron.werber@sgcowen.com

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Online links

FURTHER INFORMATION
American Society of Human Genetics: http://www.ashg.org
National Organization of Rare Diseases: http://www.rarediseases.org
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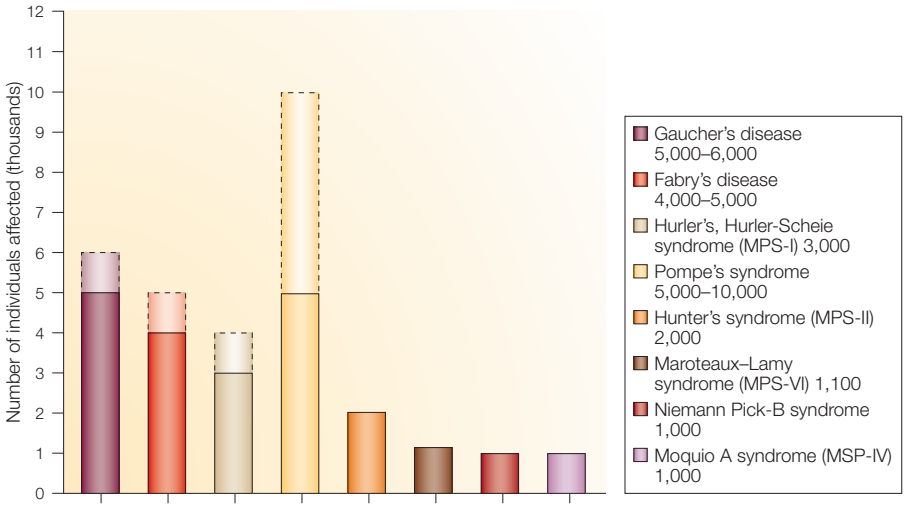


Figure 1 | **Lysosomal storage disease global prevalence.** Lysosomal storage diseases are rare inherited disorders afflicting between 1,000–10,000 patients worldwide. Given the challenge of patient diagnosis, the size of the afflicted population is directly correlated with the commercial success of the drugs.

Table 1 | **Enzyme-replacement therapies**

Disease	Drug	Company	2003 sales*	2007 sales*
Gaucher's	Cerezyme	Genzyme	732	948
Gaucher's	Zavesca	Actelion	1	27
Fabry's	Fabrazyme	Genzyme	73	397
Fabry's	Replagal	TKT	57	168
Hurler–Scheie (MPS-I)	Aldurazyme	Genzyme/BioMarin	12	204
Pompe's	Myozyme	Genzyme	N/A	65
Hunter's (MPS-II)	I2S	TKT	N/A	30
Maroteaux–Lamy (MPS-VI)	Aryplase	BioMarin	N/A	30

I2S, iduronate-2-sulphatase; N/A, not available; TKT, Transkaryotic therapies. *Estimated sales in US $ millions.

Table 2 | **Pipeline of selected therapies for lysosomal storage diseases**

Disease	Compound	Company	Mechanism	Administration	Stage
Gaucher's	Cerezyme	Genzyme	Glucocerebrosidase	30–60 U per kg IV every two to four weeks	Market
Gaucher's	Zavesca	Actelion	Glucosylceramide synthase inhibitor (small molecule)	100 mg PO TID	Market
Fabry's	Fabrazyme	Genzyme	α-Galactosidase	1 mg per kg IV qowk	Market
Fabry's	Replagal	TKT	α-Galactosidase	0.2 mg per kg IV qowk	Market (EU only)
MPS-I	Aldurazyme	Genzyme/ BioMarin	α-L-iduronidase	0.5 mg per kg IV qwk	Market
Pompe's	Myozyme	Genzyme	α-Glucosidase	20–40 mg per kg IV qowk	Phase III
Hunter's (MPS-II)	I2S	TKT	Iduronate-2-sulphatase	0.5 mg per kg IV qwk/qowk	Phase III
Maroteaux–Lamy's (MPS-VI)	Aryplase	BioMarin	Arylsulphatase B	1.0 mg per kg IV qwk	Phase III
Niemann	Pick-B NA	Genzyme	Acid sphingomyelinase	IV	Preclinical
Gaucher's	N/A	TKT	Glucocerebrosidase	IV	Preclinical
Moquio A (MSP-IV)	N/A	TKT	Galactose-6-sulphatase	IV	Preclinical
Gaucher's, Tay–Sachs, Sandhoff	GENZ-99067	Genzyme	P4 analogues	PO	Preclinical

EU, European Union; IV, intravenous; MPS, mucopolysaccharidosis; N/A, not available; PO, per oral; qowk, every other week; qwk, every week; TID, three times daily; TKT, Transkaryotic therapies.